

Where churn is most concentrated

A concise readout of the provided IBM Telco Customer Churn EDA. All figures are descriptive churn rates from the source notebook.

1. Executive Summary

Overall churn is **26.54%**. The clearest patterns sit at the start of the customer relationship and in lower-commitment arrangements. Customers in their first 0-6 months have a **53.33%** churn rate, compared with **9.51%** among customers with 49-72 months of tenure.

The source also highlights elevated churn for month-to-month customers, electronic-check users, fiber-optic customers, and customers without security or technical-support services. These are useful groups for focused testing. Because this is observational EDA, the patterns do not prove why customers leave.

2. Key Insights

01 The first six months are the sharpest risk window

Churn falls steadily across the tenure bands: **53.33%** at 0-6 months, **35.89%** at 7-12 months, and **9.51%** at 49-72 months. The source pairs this table with a density chart of tenure by churn, showing the churned-customer distribution concentrated earlier in the relationship.

02 Commitment and payment method show large gaps

Month-to-month churn is **42.71%**, versus **11.27%** for one-year plans and **2.83%** for two-year plans. Electronic-check users also show **45.29%** churn, compared with **15.24%** for automatic credit card and **16.71%** for automatic bank transfer. The source uses contract and payment count charts, plus charge-by-group boxplots, to show these differences.

03 Fiber-optic and support gaps identify a high-risk service segment

Fiber-optic customers have **41.89%** churn, versus **18.96%** for DSL. Customers without OnlineSecurity churn at **41.77%** versus **14.61%** with it; the TechSupport gap is **41.64%** versus **15.17%**. The source's service tables and security/support heatmaps make these segments easy to compare.

04 Billing and household signals extend the targeting picture

Paperless-billing customers churn at **33.57%**, versus **16.33%** for others. Senior customers churn at **41.68%** versus **23.61%** for non-seniors; customers without dependents churn at **31.28%** versus **15.45%** with dependents. The source shows demographic count plots and a payment-method-by-paperless-billing bar chart for these descriptive splits.

3. Recommendations

- 1. Build an early-life retention journey.** Focus onboarding, clear setup guidance, and proactive check-ins on the first 0-6 months. Compare results with a similar holdout group before scaling.
- 2. Test simple commitment and payment offers.** Run small, transparent tests for month-to-month and electronic-check customers, such as autopay guidance or voluntary plan-conversion offers. Track adoption, churn, and customer value.
- 3. Prioritize service-support experiments for high-risk groups.** Test help, security, and technical-support bundles for relevant fiber-optic customers. Treat service signals as a way to target experiments, not proof that an add-on will reduce churn.

Reading note: percentages summarize groups in the supplied EDA. They are associations, not causal effects.